

CN LAB – EXP-14

Nmap Network Scanning – Process

1. Aim

To perform a basic network scan using Nmap and identify accessible ports and services on an authorized target system.

2. What is Nmap?

Nmap (Network Mapper) is a network scanning tool used to discover hosts, identify open ports, and obtain basic information about network services. It is commonly used for network administration, troubleshooting, and authorized security testing.

3. Safety Note

Only scan systems or networks that you own or have explicit permission to test. For laboratory practice, use your own computer, a local virtual machine, or an authorized lab target.

4. Install and Verify Nmap

Windows: Install Nmap using the official installer.

Ubuntu/Debian:

```
sudo apt update
sudo apt install nmap
```

Verify installation:

```
nmap --version
```

5. Find the Target IP Address

On Windows, open Command Prompt and run:

```
ipconfig
```

On Linux, use:

```
ip addr
```

6. Run a Basic Nmap Scan

Replace the example IP address with the IP address of your authorized lab target:

```
nmap 192.168.1.10
```

The command performs a basic scan and reports information about accessible ports.

7. Common Nmap Commands

Purpose	Command
Basic scan	nmap 192.168.1.10

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Scan selected ports	<code>nmap -p 22,80,443 192.168.1.10</code>
Scan port range	<code>nmap -p 1-1000 192.168.1.10</code>
Service/version detection	<code>nmap -sV 192.168.1.10</code>
OS detection	<code>nmap -O 192.168.1.10</code>
Save output	<code>nmap 192.168.1.10 -oN scan.txt</code>

8. Understanding the Output

The scan output commonly contains **PORT**, **STATE**, and **SERVICE** information.

Field	Meaning
PORT	Network port number and protocol, such as 80/tcp.
STATE	Whether the port is open, closed, or filtered.
SERVICE	The service commonly associated with the port.

9. Procedure

1. Install Nmap on the system.
2. Open Command Prompt, PowerShell, or Terminal.
3. Identify the IP address of the authorized target.
4. Run **nmap TARGET_IP**.
5. Observe the discovered ports and services.
6. If required, run **nmap -sV TARGET_IP** for service/version detection.
7. Save the result using **-oN scan.txt**.
8. Document the scan findings.

10. Example

Command:

```
nmap -sV 192.168.1.10
```

This performs service/version detection against the specified authorized target. Actual results depend on the target configuration and network conditions.

11. Result

The Nmap scan was successfully performed on the authorized target, and the available port and service information was identified and documented.

12. Quick Flow

Install Nmap → Identify Target IP → Open Terminal → Run Nmap Scan → Analyze Ports/Services → Save Results → Document Findings